OTA-C 50/60 Hz Notch Filter in Sky130

Current-division OTA style: design, sizing, verification, power, timing, OTA layout and post-layout analysis

This is the third of three OTA styles. The filter, capacitors, buffer and testbench are shared with the other two styles and are derived in the current-cancellation report; this document covers what is new. Every number comes from a simulation in this project or a calculation shown here.

1. Result in one page

Quantity	Result	Target	Status
Notch frequency (tuned bias)	49.99 Hz at Ib = 0.0897 nA; 59.99 Hz at Ib = 0.1112 nA	50 / 60 Hz within 1 Hz	Met
Notch depth (vs 10 Hz gain)	22.5 dB at 50 Hz; 25.1 dB at 60 Hz	≥ 20 dB	Met, smallest margin of the three
Low-frequency gain	-6.59 dB	about −6 to −7 dB	Met
Gain above the notch	-10.83 dB at 100 Hz, -12.02 dB at 150 Hz	Not specified	Reported
Stability (2 mV step)	Settles in 41.3 ms, 14.5% overshoot	No oscillation	Met
Mains rejection in time	50 Hz is 22.5 dB below 10 Hz	≥ 20 dB	Met
THD at 5 mV / 20 mV, 10 Hz	0.69% / 2.67% at the output	≤ 2% at 5 mV	Met
Mismatch Monte Carlo (100 runs)	**47/100** keep a notch ≥ 20 dB; 39 have no notch at all	Reported	**Poor yield**
Power	1.549 µW total: filter 12.33 nW, buffer 1.565 µW	Reported	Buffer dominates
OTA layout	294.9 × 23.6 µm; DRC 0 errors; LVS: circuits match uniquely	Clean	Met
Post-layout	49.98 Hz, 21.0 dB at Ib = 0.0927 nA; Monte Carlo 33/100 usable	Close to schematic	Met

Honest summary: this style needed four design cycles and ends with the least margin: the notch is 22.5 dB deep against a 20 dB target, because the OTA's open-loop gain is only 25.5 dB. It also has the largest systematic offset relative to its linear range and the largest layout.

2. How the current-division OTA works

Each input drives three PMOS source followers in parallel, M_C, M_B and M_A, sized $N_C : 1 : N_A$ units. They share a source node (s1 or s2) fed by a current source, so that node follows its input one gate voltage higher. A transistor M_R biased in its linear region connects s1 and s2 through a common centre node, and turns the input difference into a current $g_R \cdot v_d$.

That current has to come out of the followers, and it divides between them in proportion to their size. M_C's share is dumped to VSS (current division). M_B and M_A feed the output with opposite signs through the mirror M3/M4 (current cancellation):

$$I_{out} = [i(M_{A1}) + i(M_{B2})] - [i(M_{B1}) + i(M_{A2})]$$

$$g_m = [(N_A - 1)/(N_A + 1 + N_C)] \cdot g_{R,eff},\ 1/g_{R,eff} = 1/g_R + 2/g_f$$

where g_f is the total transconductance of one side's followers. Both division and cancellation shrink g_m, which is how the paper reaches nA/V levels.

Tuning. M_R's gate comes from the diode M6, whose current is set by M7. The paper drives M7 with a tuning voltage. Here M7 mirrors a tuning current I_b, and M5 supplies $I_b/2$ to the centre node. In weak inversion M_R's conductance is then proportional to I_b, so f_0 scales with I_b as in the other two styles and the same tuning flow applies.

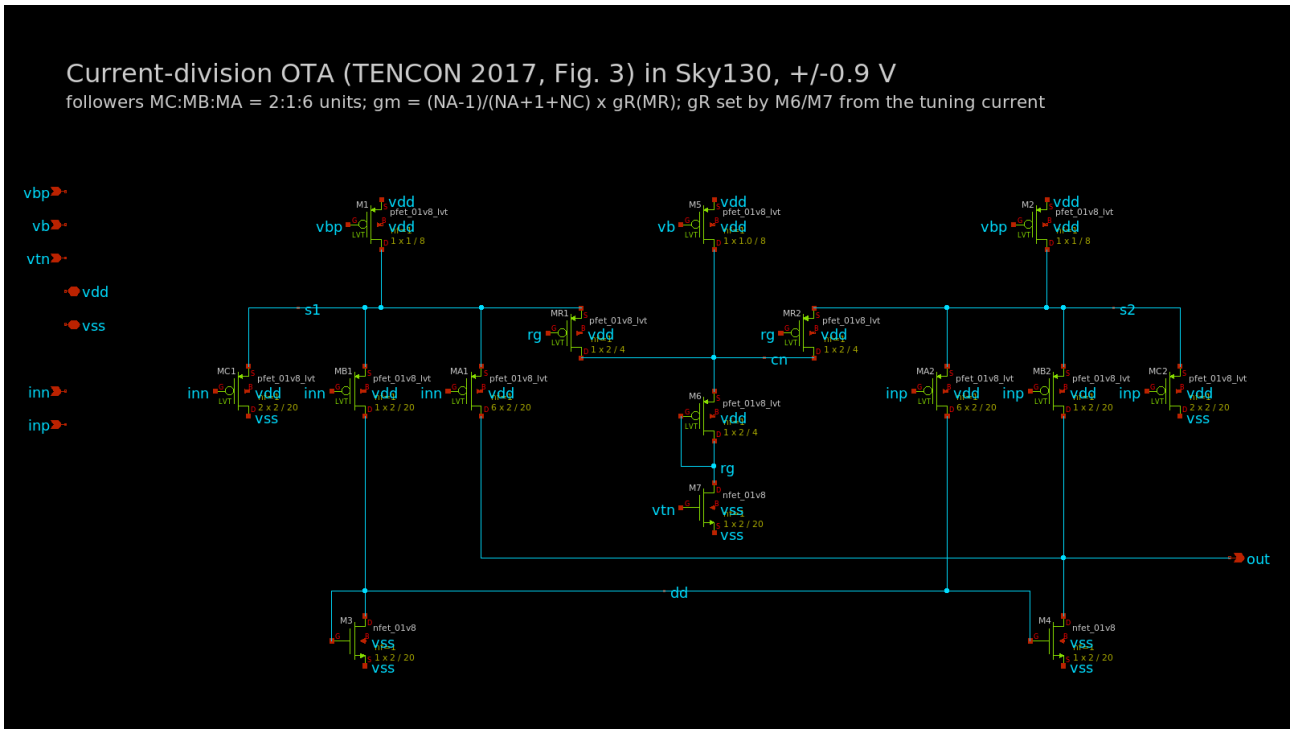


Figure 1. Current-division OTA in Xschem.

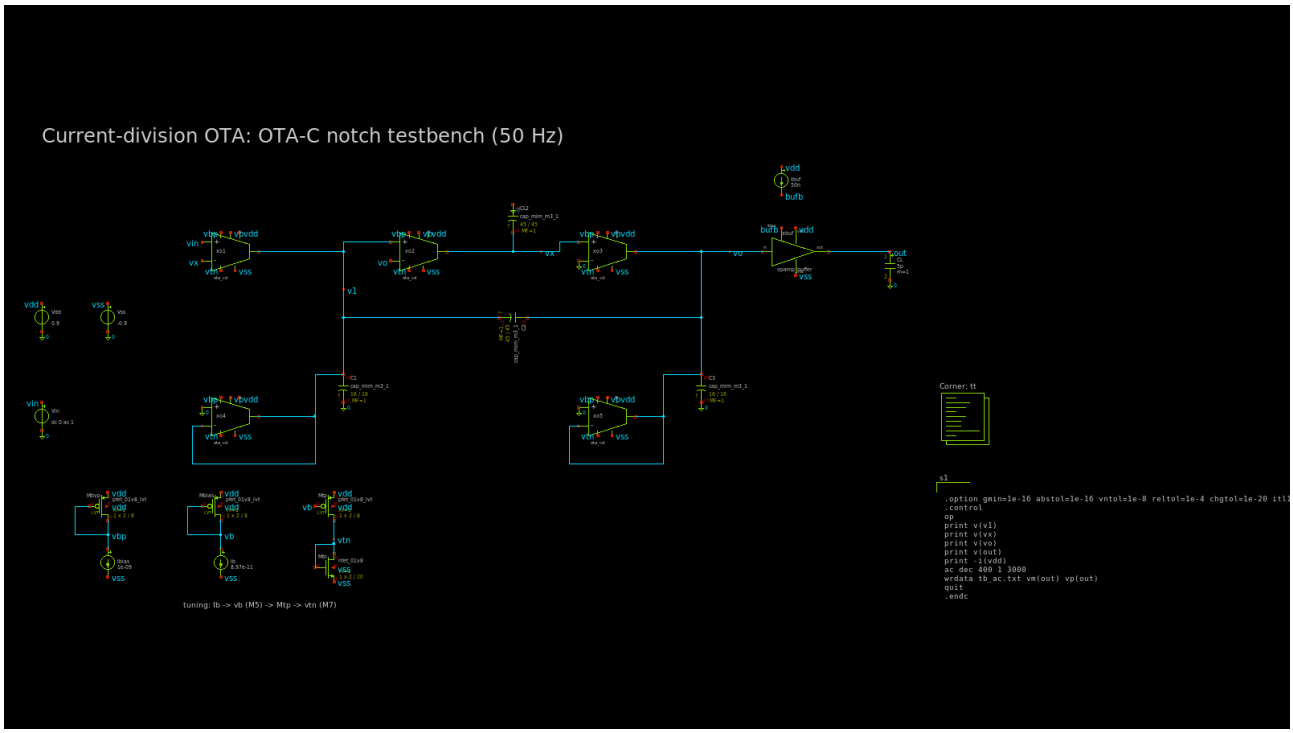


Figure 2. Notch testbench with five current-division OTAs, the follower bias, the tuning chain and the buffer.

3. Four design cycles

Cycle	Sizes	Finding	Change
1	N_C:1:N_A = 6:1:2, 3 nA/side, L = 4/8 µm	Offset 47 mV (≈ the whole ±46 mV range), gain 20 dB	Longer channels
2	Same with L = 20 µm, then 1 nA/side	Offset only 35 → 26 mV: not channel-length modulation, but the output and mirror nodes sitting at different DC voltages	Accept offset, test the filter
3	6:1:2 at 1 nA/side	Notch could not reach 50 Hz: f_0 peaked at 47 Hz. g_m cannot exceed $(N_A-1)/(N_A+1+N_C)\cdot g_f/2$ even with M_R shorted	2 nA/side, N_C = 4 to lift the ceiling
3b	4:1:2 at 2 nA/side	Tunes to 50 and 60 Hz, but depth 15.8 / 18.4 dB because A_0 = 22.8 dB	Reduce the output-branch current
4 (final)	2:1:6 at 0.5 nA/side	Depth 22.5 / 25.1 dB, A_0 = 25.5 dB	Frozen

The key step in cycle 4 is a scaling argument. For a fixed target g_m with some headroom ($g_{m,max}$ = 2 g_m), the output-branch current works out to

$$I_{\text{out branch}} \approx 4\, g_m\, (N_A + 1) \,/\, [(N_A - 1)\,(g_m/I_D)]$$

This is independent of N_C, and it falls as N_A grows. Output resistance, and so open-loop gain, rise as that current falls, and the systematic offset shrinks with it. So the final design uses a large N_A = 6 and a small N_C = 2, the opposite proportions from the paper's 2.1 and 22.

Final OTA at I_b = 0.0897 nA: g_m 1.511 nA/V, linear range -31 to 33 mV, systematic offset 26.1 mV, R_{out} 12.4 GΩ, A_0 25.5 dB, CMRR 46.2 dB, input noise 334 µV rms, OTA plus bias 4.74 nW.

The offset comes from the output node sitting at 0 V while the mirror node sits about 0.36 V above VSS: the devices on the two nodes see different drain voltages, and longer channels barely changed it. A cascode or a common-mode-matched mirror would be the fix, but that would change the paper's topology.

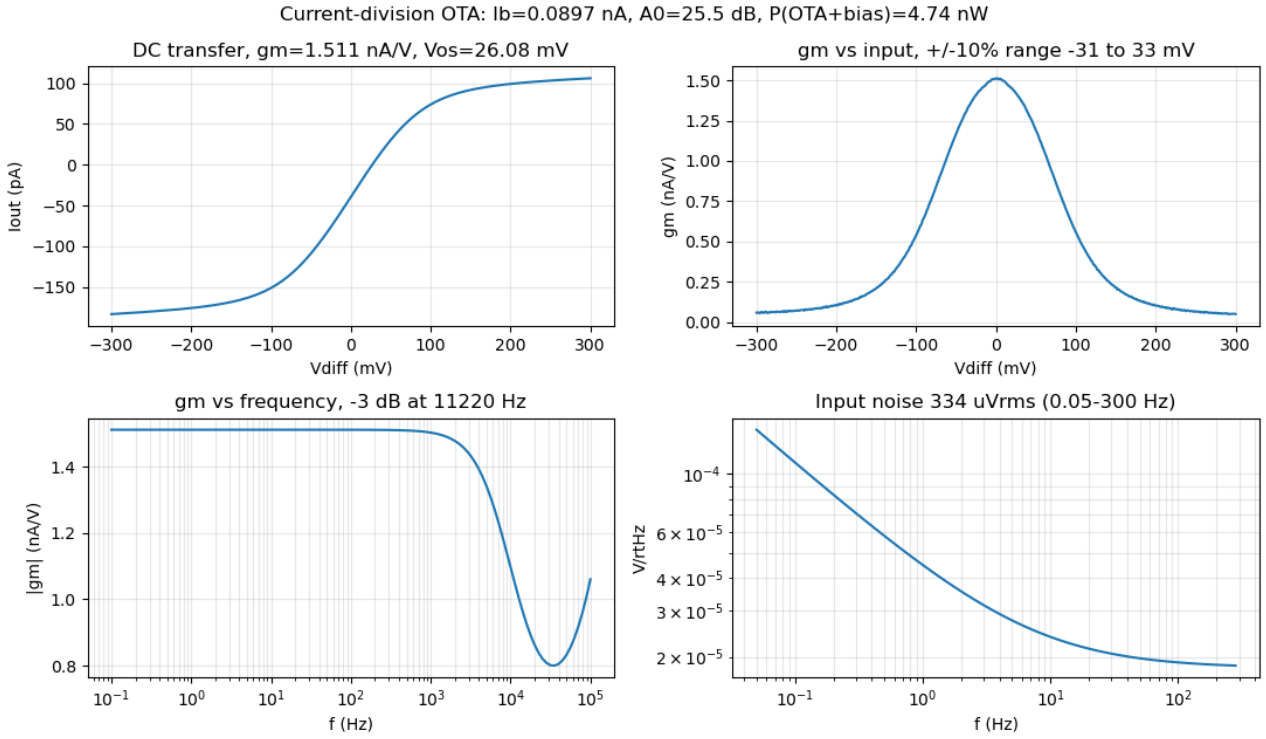


Figure 3. OTA validation: DC transfer, g_m vs input, g_m vs frequency, input noise.

4. Filter results

Frequency (Hz)	1	10	25	40	75	100	150	300
Gain (dB)	-6.57	-6.59	-6.80	-9.78	-11.18	-10.83	-12.02	-16.17

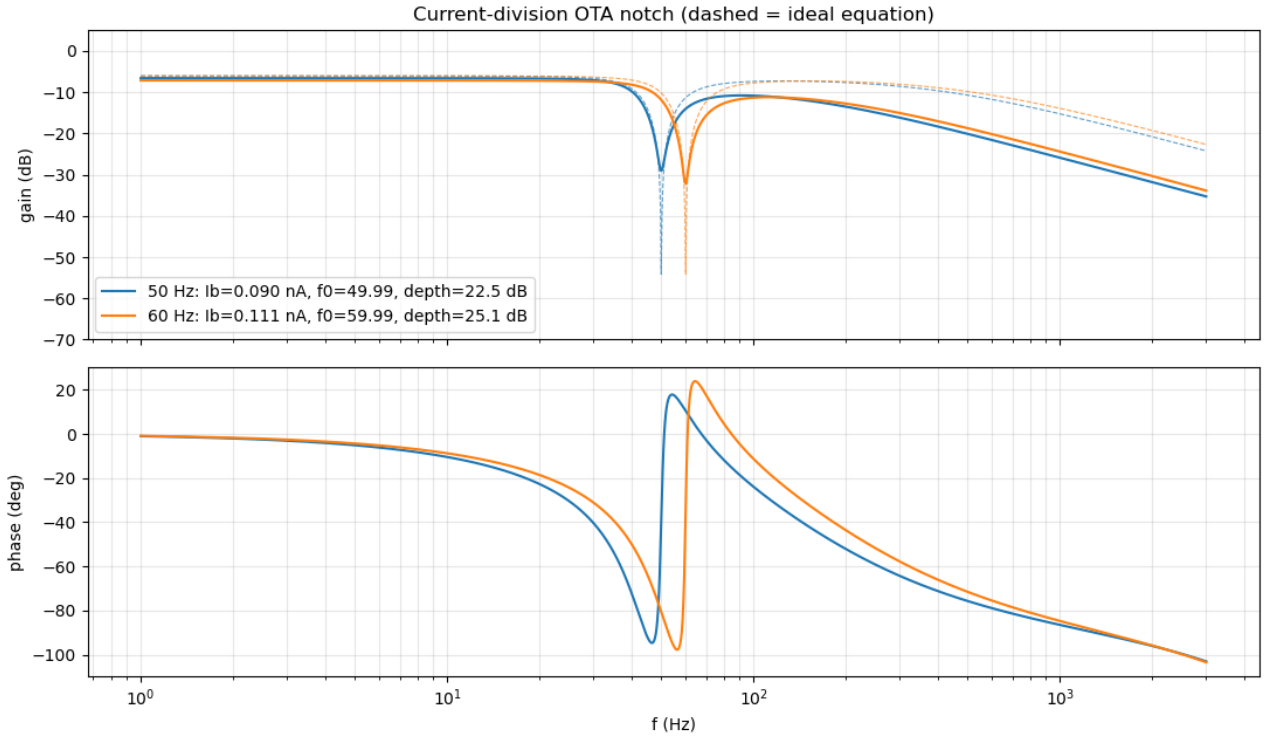


Figure 4. Notch response at the buffer output, tuned to 50 Hz and 60 Hz.

Transient. A 2 mV step settles to 0.929 mV in 41.3 ms with 14.5% overshoot, and does not oscillate. In the 10 + 50 Hz mix the 50 Hz tone comes out 22.5 dB below the 10 Hz tone.

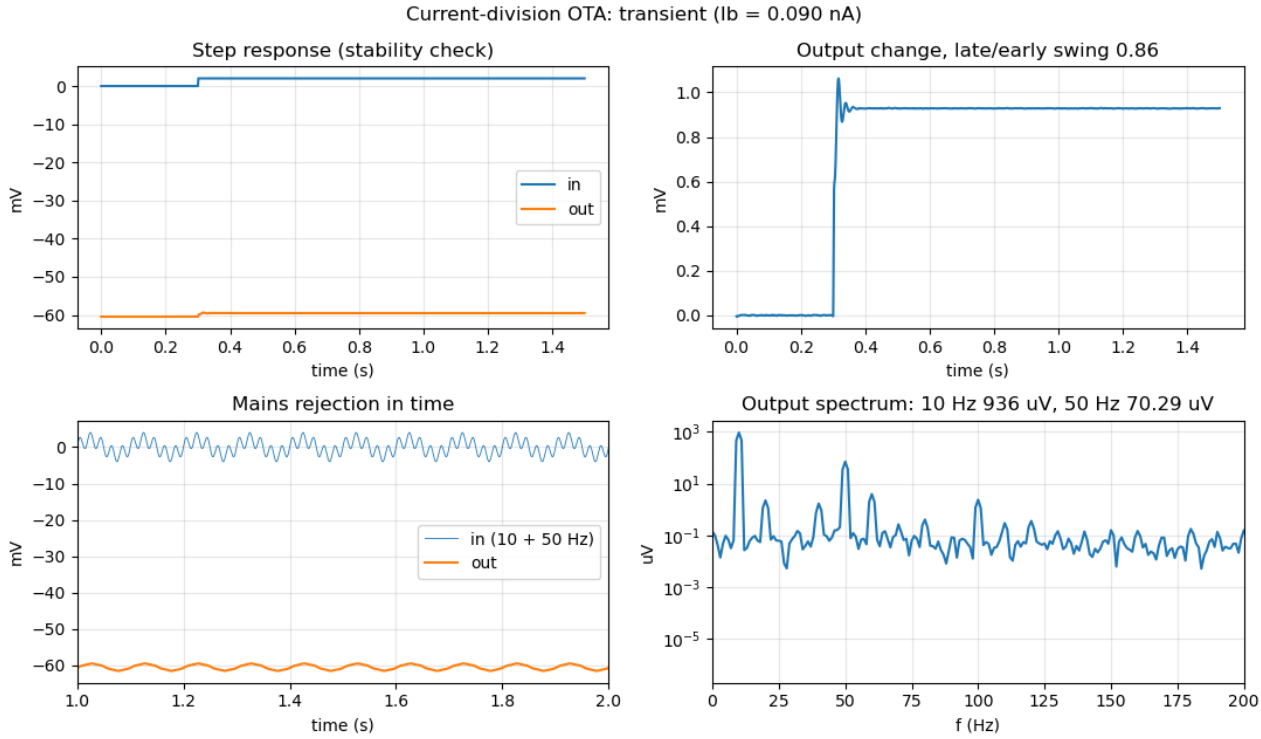


Figure 5. Step response and mains-rejection test.

5. Monte Carlo, corners and distortion

Mismatch Monte Carlo, 100 runs. A run is scored as a working notch only if its notch lies between 20 and 200 Hz, its depth is at least 20 dB and its 10 Hz gain is within 6 dB of nominal. **47 of 100 runs pass.** 39 runs have no notch at all: device offsets drive an OTA out of its -31 to +33 mV linear range and the filter's DC point collapses (output near −0.65 V or +0.27 V). Of the passing runs, f_0 = 48.51 ± 10.06 Hz and depth 23.5 ± 2.2 dB; overall 11% of runs land within ±2 Hz and 19% within ±5 Hz.

This is the decisive weakness of this style in Sky130: the nominal design meets the targets, but it does not survive mismatch. The depth margin (about 2.5 dB) and the offset (26.1 mV, about 79% of the linear range) both trace back to the OTA's 25.5 dB open-loop gain and its unmatched output and mirror nodes. A cascoded output mirror would raise the gain and reduce the offset, at the cost of departing from the paper's circuit.

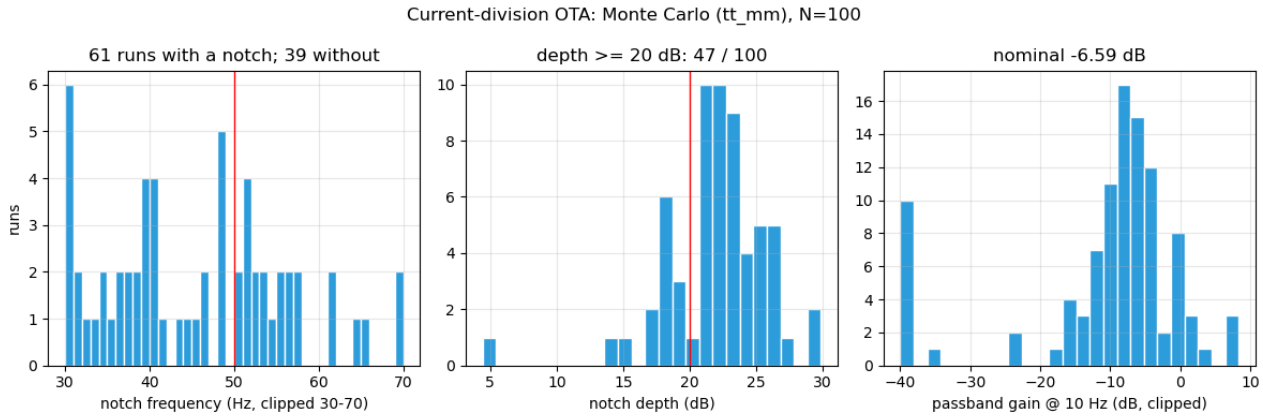


Figure 6. Monte Carlo histograms (mismatch, 100 runs).

Corner	T (°C)	f_0 at fixed Ib (Hz)	Depth (dB)	Gain 10 Hz (dB)	Ib trimmed (nA)	f_0 trimmed (Hz)	Depth trimmed (dB)
tt	27	49.99	22.5	-6.59	0.0897	50.00	22.5
ss	27	47.87	21.9	4.76	0.0937	49.77	22.4
ff	27	53.27	22.0	-7.21	0.0842	50.53	21.2
sf	27	48.46	23.0	-6.95	0.0925	49.81	23.4
fs	27	60.34	22.1	-7.98	0.0743	53.62	20.5
tt	37	49.36	22.3	-6.98	0.0909	49.92	22.5
tt	0	51.95	22.7	-1.50	0.0863	50.22	22.2
tt	85	50.85	21.4	-7.71	0.0882	50.25	21.3

Every corner keeps 20–23 dB of depth. One proportional trim step brings f_0 within 0.6 Hz of 50 Hz except at fs, which stops at 53.6 Hz after the single step and would need a second.

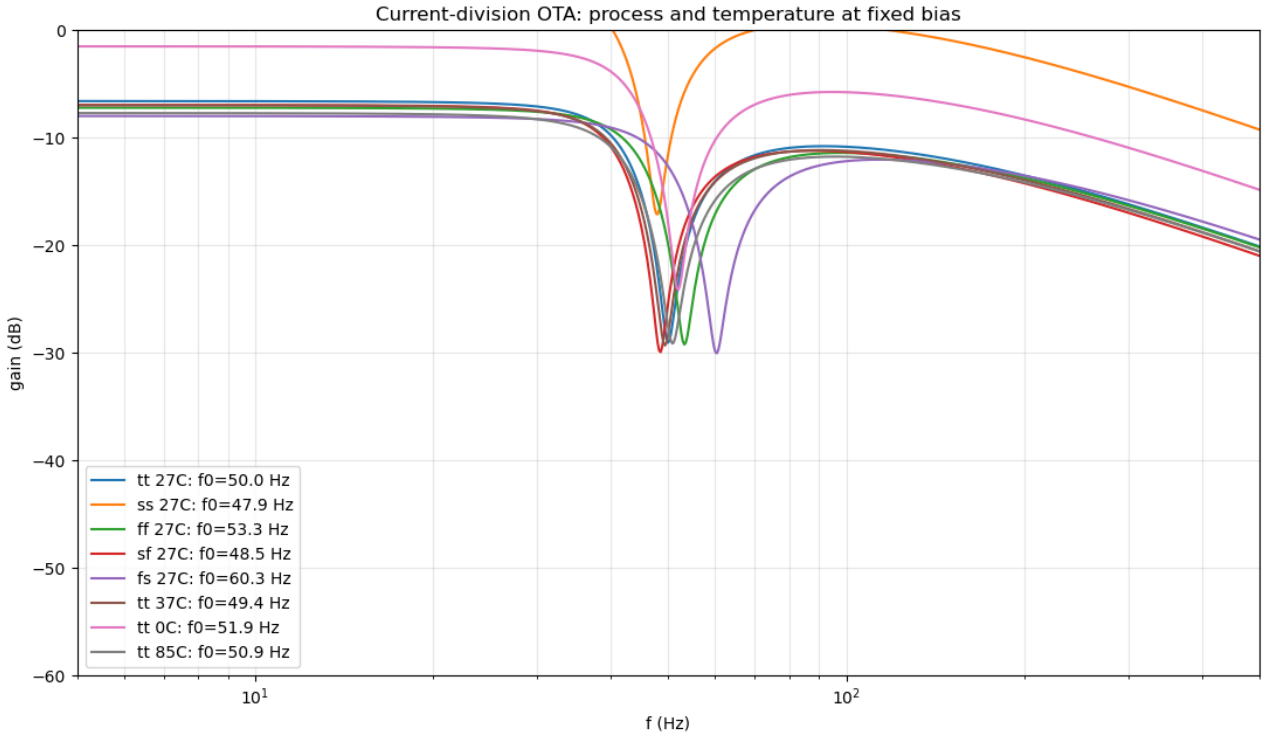


Figure 7. Response across process corners and temperature at fixed bias.

Input (mV peak)	Output fundamental (mV)	THD at output (%)	THD at filter output (%)	HD2 (%)	HD3 (%)
5.0	2.341	0.686	0.707	0.684	0.005
10.0	4.686	1.343	1.386	1.343	0.040
20.0	9.406	2.673	2.754	2.668	0.163

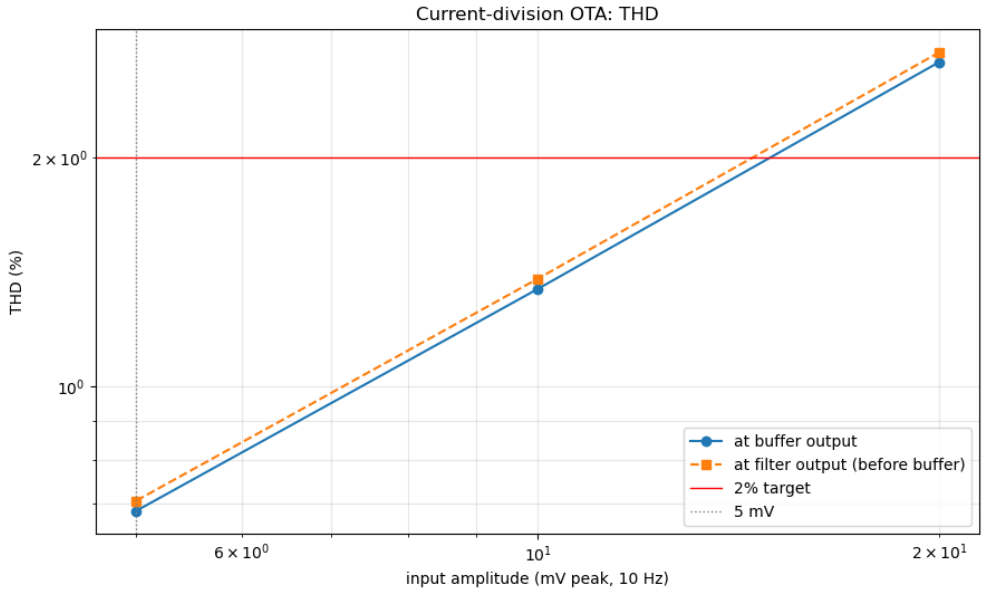


Figure 8. THD vs input amplitude.

6. Power

$I_{filter} = 5 \times I_{OTA}$ + bias branches = 6.85 nA, $P_{filter} = 1.8\ V \times 6.85\ nA =$ **12.33 nW**

$P_{total} = 1.8\ V \times 860.8\ nA$ (full testbench) = **1.549 µW**

Each OTA draws two 0.5 nA follower currents plus the centre-node and tuning branches, and the bias network adds a fixed 1 nA follower reference and the tuning chain. This style uses the most current of the three, because the followers must carry far more current than the output to keep g_f well above g_R.

7. Timing

Quantity	Value
Group delay	2.84 ms at 1 Hz; 3.05 ms at 10 Hz; 3.89 ms at 20 Hz; 6.33 ms at 30 Hz
Step rise 10–90%	10.08 ms
Settling to ±2%	41.3 ms
Overshoot	14.5%
OTA maximum output current	389 pA (output-branch current of one side)
Slew rate on C_{L2}	95.3 V/s
Slew-limited amplitude at 50 Hz	303 mV peak

8. Layout

All PMOS share one n-well tied to VDD. Row 1 holds side 1 (M1, the nine follower units in the order C A A A B A A A C, MR1, M6, M5) with gates facing down into a metal-1 band of 13 buses. Row 2 holds side 2 (M2, its nine followers, MR2) with gates facing up into the same band. The three NMOS devices sit outside the well with their own substrate tap.

Because the two rows send vertical straps into the same band from opposite sides, the generator checks every pair of vertical straps and shifts row 2 until none collide. The first pass had 12 DRC errors: two contacts 0.18 µm apart on the same bus, below the 0.19 µm contact rule, because the check used the 0.17 µm interconnect rule. With the check at 0.2 µm the result is **DRC 0 errors, LVS circuits match uniquely** (27 extracted transistors, merged into the 15 schematic devices). The layout is 294.9 × 23.6 µm, about 6,960 µm², the largest of the three, because it has 24 PMOS units in a single row pair.

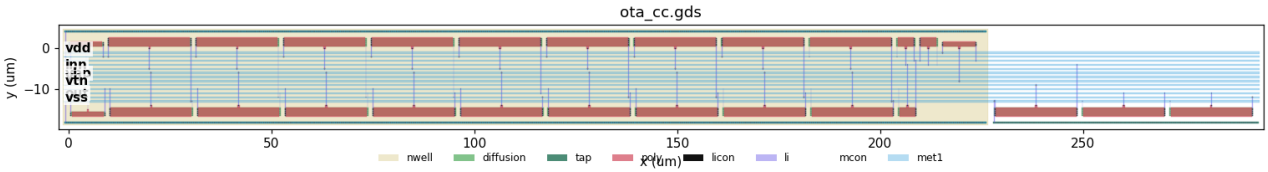


Figure 9. Current-division OTA layout rendered from ota_cd.gds.

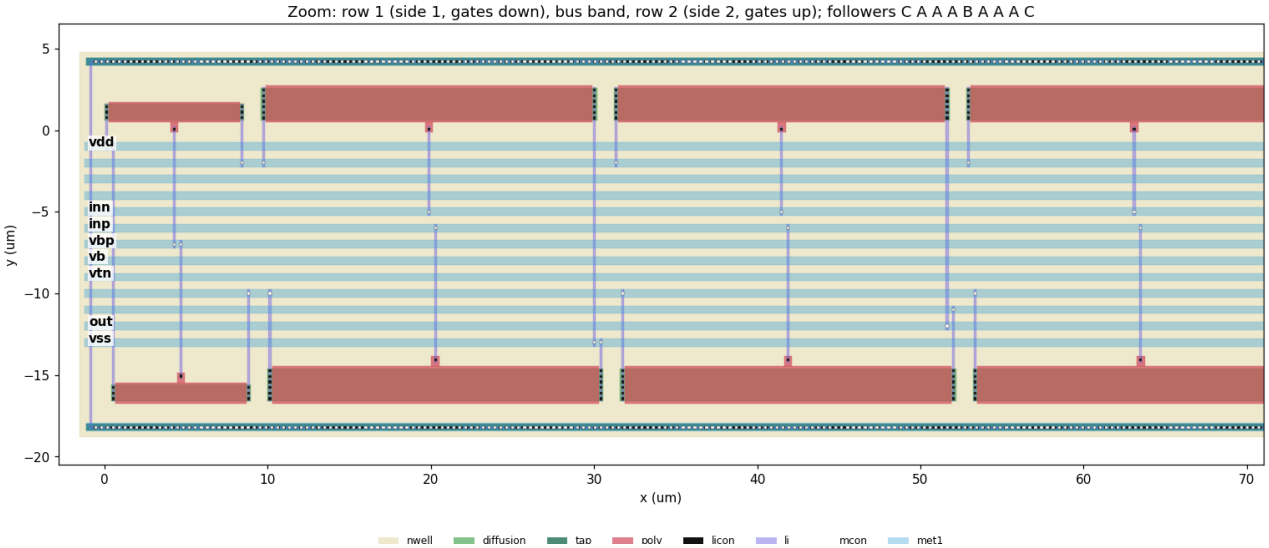


Figure 10. Zoom on the left end: row 1, bus band, row 2.

9. Post-layout analysis

Quantity	Schematic	Extracted layout
Bias for 50 Hz	0.0897 nA	0.0927 nA (+3.3%)
Bias for 60 Hz	0.1112 nA	0.1152 nA
Notch depth 50 / 60 Hz	22.5 / 25.1 dB	21.0 / 23.4 dB
Gain at 10 / 100 / 150 Hz	-6.59 / -10.83 / -12.02 dB	-6.70 / -11.59 / -12.94 dB
THD at 5 / 10 / 20 mV	0.69 / 1.34 / 2.67 %	0.62 / 1.24 / 2.50 %
Monte Carlo: working notch ≥ 20 dB	47 / 100	33 / 100
Monte Carlo: no notch at all	39 / 100	43 / 100
Monte Carlo f_0 (working runs)	48.51 ± 10.06 Hz	50.53 ± 7.40 Hz
Monte Carlo depth (working runs)	23.5 ± 2.2 dB	22.5 ± 1.6 dB

The post-layout step settles to 0.921 mV with 14.0% overshoot; after settling, the remaining ripple is 0.5–1.7 µV peak-to-peak and does not grow. During post-layout tuning at 60 Hz, one operating-point solve landed on a latched state (output +286 mV, no notch). Repeating the same bias with and without an initial-condition hint gave the normal state every time, so it was a one-off convergence choice by the simulator, not a reproducible second state.

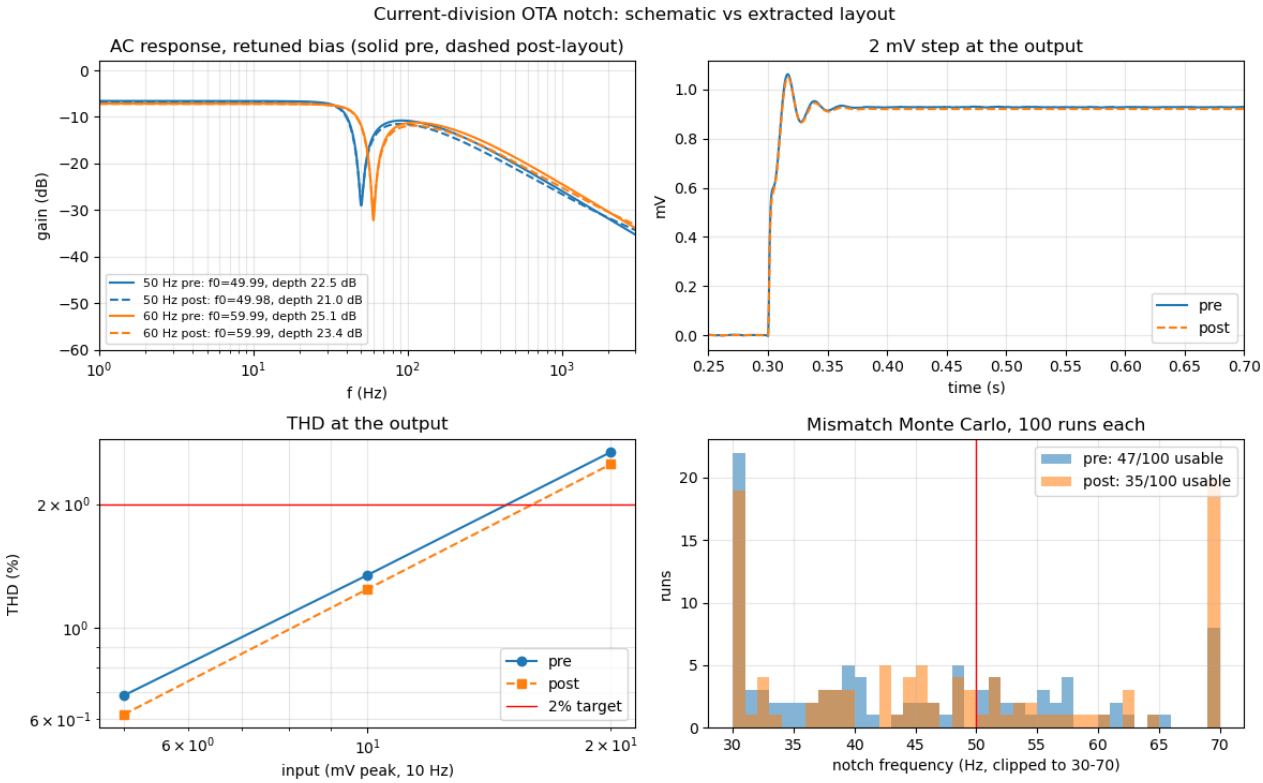


Figure 11. Schematic (solid) vs extracted layout (dashed).

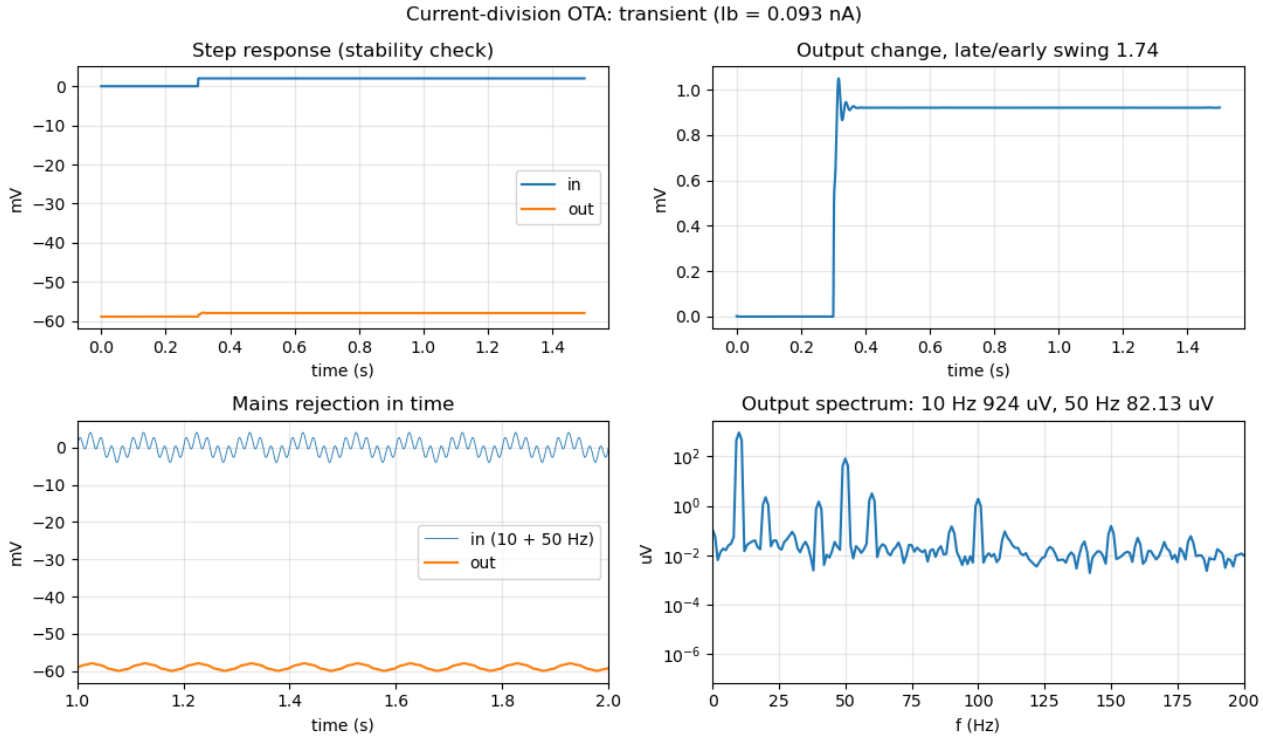


Figure 12. Post-layout transient.

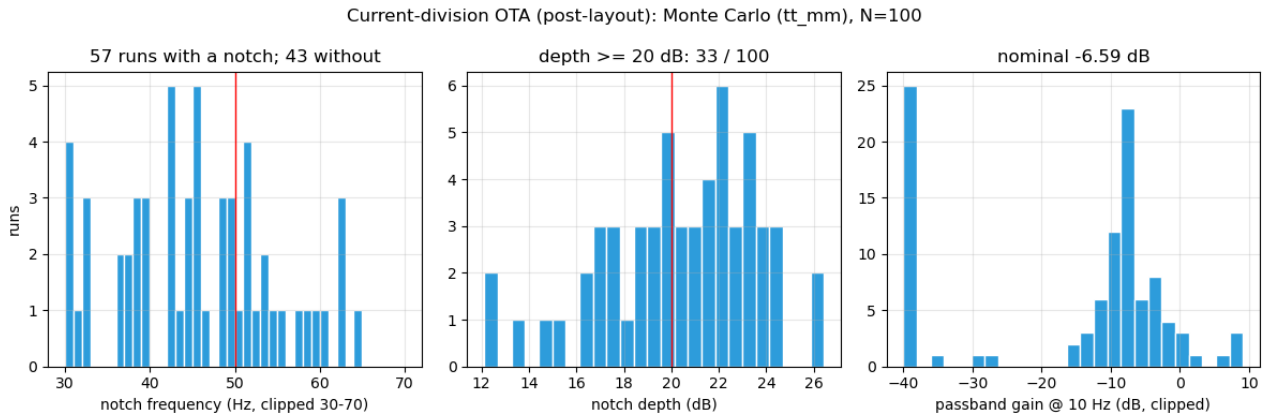


Figure 13. Post-layout Monte Carlo (100 runs, mismatch).

10. Limitations

• **Mismatch yield:** only 47% of Monte Carlo runs keep a working notch (post-layout 33%), against 93–100% for the other two styles.

• **Small depth margin:** 22.5 dB against a 20 dB target, set by the OTA's 25.5 dB gain.

• **Systematic offset** of 26.1 mV is about 79% of the linear range; fixing it would need a cascoded or level-matched mirror, which changes the paper's topology.

• **Sizing departs from the paper:** N_A = 6 and N_C = 2 instead of about 2 and 22, chosen by the scaling argument in section 3.

• **Shared limits:** corners instead of process Monte Carlo, an ideal bias reference, only the OTA laid out, no dummy devices, and the shared buffer's distortion.